

Clinical features

Dermatitis herpetiformis presents with diffuse, symmetrical, grouped polymorphic lesions including erythema, urticarial plaques, papules, herpetiform vesicles, and blisters, which progress to erosions, excoriations, and hyperpigmentation. The most commonly affected sites are the extensor surfaces of the elbows (90%), knees (30%), shoulders, buttocks, sacral region, and face. Pruritus of variable intensity, burning sensations, and scratching preceding lesion development are common.

Although small bowel involvement is often asymptomatic in adults, it may manifest with abdominal pain, diarrhea, iron deficiency anemia, and in children, reduced growth velocity.

The clinical presentation of dermatitis herpetiformis is frequently atypical, leading to potential diagnostic delay. In children, the main differential diagnoses include atopic dermatitis, scabies, papular urticaria, and impetigo. In adults, the differential diagnosis should include eczema, other autoimmune blistering diseases—particularly linear IgA disease and bullous pemphigoid—nodular prurigo, urticaria, and polymorphic erythema. The diagnosis is established through clinical evaluation, histopathology, and especially direct immunofluorescence (DIF).

Differential diagnosis of dermatitis herpetiformis

Disease	Clinical picture	Clinical localization	Histopathology	Immunopathology
Dermatitis herpetiformis	Polymorphic lesions, herpetiform vesicles	Extensor limbs, sacral region, buttocks	Subepidermal blister with neutrophils at papillary tips	Granular IgA at basement membrane zone
Linear IgA dermatitis	Herpetiform bullae and vesicles	Periorificial regions	Subepidermal blisters with neutrophil infiltration	Linear IgA at basement membrane zone
Bullous pemphigoid	Large, grouped, tense blisters	Proximal limbs, lower abdomen	Subepidermal blisters with eosinophil infiltration	Linear IgG at basement membrane zone
Papular urticaria	Pruritic papules and wheals	Limbs, trunk	Dermal edema, perivascular infiltrate with eosinophils	Non-specific findings
Atopic dermatitis	Eczematous, pruritic lesions	Face, flexural areas (varies with age)	Acanthosis, spongiosis, lymphomonocytic infiltrate	Non-specific findings
Scabies	Burrows and polymorphic pruritic lesions	Interdigital areas, axillae, genitalia, buttocks	Mixed perivascular infiltrate, occasional mite detection	Non-specific findings
Urticaria	Pruritic wheals, angioedema	Diffuse	Dermal edema, perivascular lymphomonocytic infiltrate	Non-specific (except in urticarial vasculitis)

NG: neutrophil granulocytes; EG: eosinophil granulocytes; BMZ: basement membrane zone

Histopathology

The characteristic histopathological findings in lesional skin are subepidermal blisters with accumulation of neutrophils (and few eosinophils) at the tips of dermal papillae. While these features may be suggestive, histopathology alone is not diagnostic. A non-specific histological pattern is frequently observed, emphasizing the need for direct immunofluorescence to confirm the diagnosis.